

Nexivarin 10 mg Tablets — eCTD Submission
Module 3 — Quality

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
Submission Date	2021-03-15	Section	3.2.S.1
Prepared by	PharmaCorp Inc.	Confidentiality	Proprietary and Confidential

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PharmaCorp Inc.
NDA 123456

Nexivarin 10 mg Tablets

Section 3.2.S.1
Drug Substance — General Information

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3.2.S.1 General Information — Nexivarin Hydrochloride

3.2.S.1.1 Nomenclature

Type	Name
INN / Recommended Name	Nexivarin
Chemical Name (IUPAC)	(RS)-3-Ethyl 5-methyl 2-[(2-aminoethoxy)methyl]-4-(2-chlorophenyl)-1,4-dihydropyridine-3,5-dicarboxylate hydrochloride
Chemical Name (CAS)	3-Ethyl 5-methyl (+)-2-[(2-aminoethoxy)methyl]-4-(2-chlorophenyl)-1,4-dihydro-6-methyl-3,5-pyridinedicarboxylate, monohydrochloride
CAS Registry Number	127620-12-7 (hypothetical)
Molecular Formula	C ₂₀ H ₂₅ ClN ₂ O ₅ ·HCl
Molecular Weight	479.35 g/mol
Drug Master File (DMF)	DMF #32456 (US FDA)
ATC Code	C08CA (Dihydropyridine calcium channel blockers)

3.2.S.1.2 Structure

The chemical structure of nexivarin hydrochloride is a racemic mixture of the (R) and (S) enantiomers. The dihydropyridine ring system is the pharmacophore responsible for calcium channel blocking activity. The structure has been confirmed by:

- ¹H-NMR (400 MHz, DMSO-d₆): δ 9.12 (s, 1H, NH), 8.03 (d, J=7.8 Hz, 1H, ArH), 7.52–7.34 (m, 3H, ArH), 5.38 (s, 1H, C₄-H), 4.02–3.88 (m, 4H, OCH₂CH₂), 3.75 (s, 3H, OCH₃), 3.72 (q, J=7.1 Hz, 2H, OCH₂CH₃), 2.98 (t, J=5.8 Hz, 2H, NCH₂), 1.18 (t, J=7.1 Hz, 3H, CH₃)
- ¹³C-NMR, DEPT, COSY, HMBC: All carbons, proton assignments, and long-range correlations are consistent with proposed structure
- HRMS (ESI⁺): m/z found 443.1524 [M+H]⁺ (calc. 443.1530 for C₂₀H₂₆ClN₂O₅⁺)
- IR (KBr): 3310 (N-H), 2940 (C-H), 1700, 1680 (C=O), 1490 (C=C), 750 cm⁻¹ (Ar-Cl)
- XRPD: Crystalline Form I (stable polymorph) confirmed by characteristic peaks at 2θ = 8.2°, 12.5°, 15.8°, 18.4°, 22.1°, 25.3°

3.2.S.1.3 General Properties

Property	Value	Method
Physical Form	White to off-white crystalline powder	Visual
Odour	Odourless	Organoleptic
Taste	Slightly bitter	Organoleptic (trained panel)
Melting Point	196–200°C (dec.)	DSC (10°C/min)
Optical Rotation	[α] _{D25} = 0° (racemic)	Polarimetry, c=1 in MeOH
Solubility in Water (25°C)	2.8 mg/mL (slightly soluble)	HPLC assay
Solubility in 0.1N HCl (25°C)	18.4 mg/mL (freely soluble)	HPLC assay
Solubility in pH 6.8 PBS	0.42 mg/mL (slightly soluble)	HPLC assay
Solubility in Ethanol (25°C)	46.2 mg/mL (freely soluble)	HPLC assay
log P (n-octanol/water)	3.05 (experimental)	Shake-flask

Property	Value	Method
pKa (basic)	9.4	Potentiometric titration
BCS Classification	Class II	High permeability, low solubility
Hygroscopicity	Non-hygroscopic (< 0.2% wt gain to 90% RH)	DVS
Polymorphism	Form I (stable); Form II (metastable)	XRPD, DSC